# Practice workbook · Explain how insulin and glucagon control blood glucose

Pancreatic hormones, liver glycogen storage and release, and completing glucose control diagrams, including naming the hormones and glucose as the blood sugar.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BLOOD-GLUCOSE-INSULIN-GLUCAGON

# 1. Question 8 · part a · 3 marks

June 2024 Paper 2R

Worked-example retry · Attempt part a only.

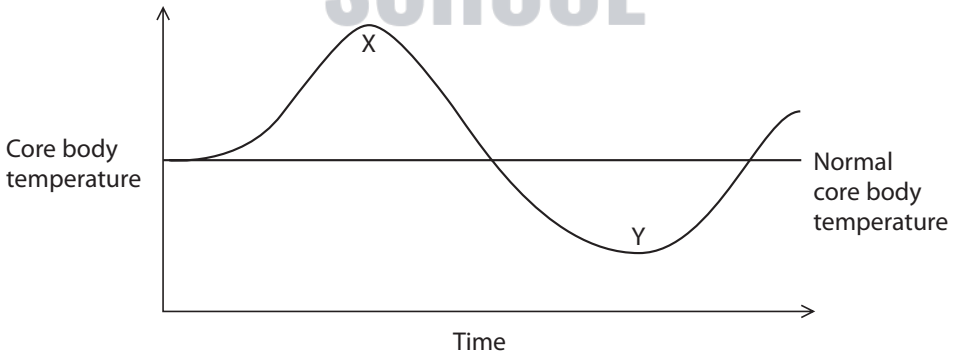
**8** (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

**(3)**

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.




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## 1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 7 · part b.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

**7** The table gives the percentage of overweight people in different age groups for six countries.

| Country | Percentage overweight | | Percentage overweight for each age group | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Males | Females | 18–24 | 25–34 | 35–44 | 45–64 | 65–74 | 75 years and over |
| Croatia | 67.5 | 48.3 | 22.3 | 38.1 | 53.7 | 69.2 | 73.1 | 65.7 |
| Malta | 66.8 | 55.2 | 36.0 | 52.5 | 62.7 | 69.0 | 74.1 | 65.3 |
| Poland | 64.1 | 46.7 | 19.3 | 38.7 | 52.8 | 66.0 | 73.2 | 65.5 |
| Portugal | 57.6 | 50.0 | 23.6 | 35.1 | 49.9 | 63.5 | 69.3 | 71.2 |
| Turkey | 56.4 | 56.3 | 22.4 | 44.2 | 64.9 | 75.4 | 66.2 | 56.5 |
| UK | 60.2 | 51.8 | 29.0 | 47.3 | 56.0 | 64.0 | 62.8 | 52.6 |

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

**(2)**

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ratio = .....

(ii) Give two trends shown in the data.

**(2)**

1 .....

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2 .....

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(iii) Give two reasons why the data for a country might be unreliable.

**(2)**

1 .....

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2 .....

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## 2. Question 7 · part b.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Individuals who are overweight or obese have an increased risk of diabetes.

People with diabetes can have high blood glucose levels.

Explain how blood glucose levels are reduced in a person who does not have diabetes.

**(3)**

(ii) Suggest why one symptom of diabetes is frequent urination.

**(2)**

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**(Total for Question 7 = 11 marks)**

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## 2. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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## 3. Question 4 · part b.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.ii only.

**4** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

**(3)**

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(ii) Give two health risks, other than diabetes, associated with obesity.

**(2)**

1 .....

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2 .....

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# 3. Question 4 · part b.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.ii only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ..... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ..... million

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## 3. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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# 4. Question 4 · part b.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.i only.

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**4** Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

**(3)**

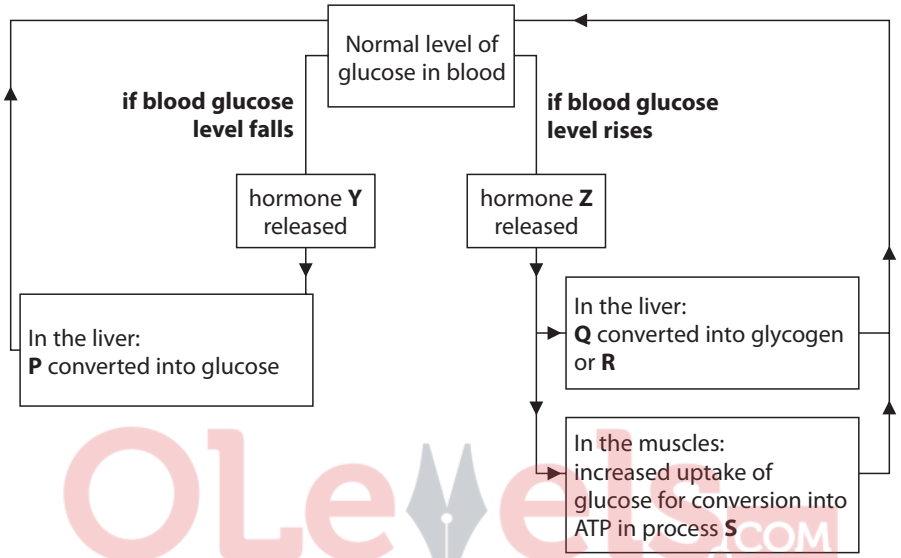
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# 4. Question 4 · part b.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.i only.

(b) The diagram shows the homeostatic control of blood glucose levels.




(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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P 6 7 0 5 9 A 0 1 2 2 4

## 4. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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## 5. Question 4 · part b.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.ii only.

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**4** Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

**(3)**

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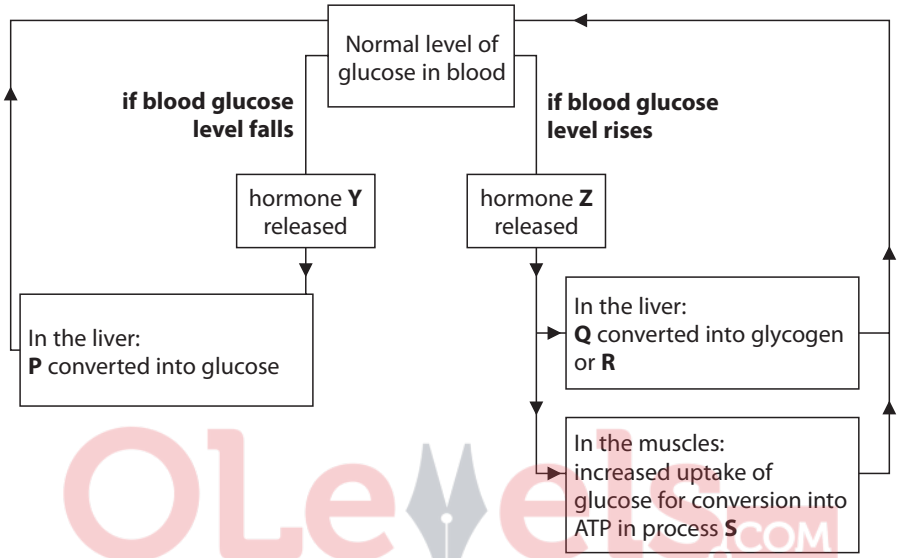
P 6 7 0 5 9 A 0 1 1 2 4

## 5. Question 4 · part b.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(b) The diagram shows the homeostatic control of blood glucose levels.




(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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12

P 6 7 0 5 9 A 0 1 2 2 4

# 5. Question 4 · part b.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(iii) Identify P, Q, R and S.

**(4)**

P .....

Q .....

R .....

S .....

(c) The homeostatic process shown in the diagram is controlled by negative feedback.

Using the example shown in the diagram, explain what is meant by negative feedback.

**(4)**

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**(Total for Question 4 = 13 marks)**

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## 5. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part b.iii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.iii only.

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**4** Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

**(3)**

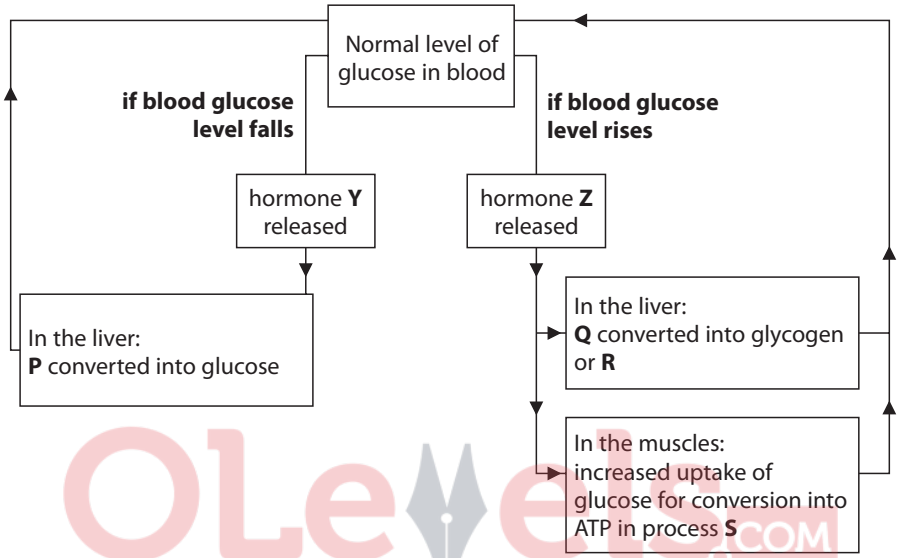
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# 6. Question 4 · part b.iii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.iii only.

(b) The diagram shows the homeostatic control of blood glucose levels.




(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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P 6 7 0 5 9 A 0 1 2 2 4

# 6. Question 4 · part b.iii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.iii only.

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(iii) Identify P, Q, R and S.

**(4)**

P .....

Q .....

R .....

S .....

(c) The homeostatic process shown in the diagram is controlled by negative feedback.

Using the example shown in the diagram, explain what is meant by negative feedback.

**(4)**

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**(Total for Question 4 = 13 marks)**

## 6. Question 4 · part b.iii · 4 marks · Working and answer

Working and answer

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# 7. Question 7 · part b · 1 mark

June 2022 Paper 1

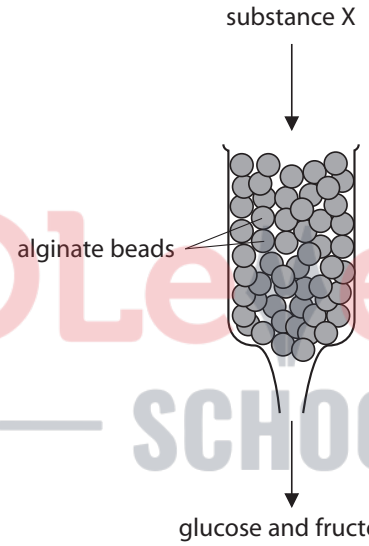
Further practice · Attempt part b only.

**7** (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

**(1)**

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.




Name substance X.

**(1)**

(iii) Describe how to prepare alginate beads.

**(3)**

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**20**

P 7 0 9 7 3 A 0 2 0 2 4

# 7. Question 7 · part b · 1 mark

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Further practice · Attempt part b only.

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(b) Which process is involved in the conversion of glucose into glycogen in the human body?

**(1)**

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

**(2)**

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number of people = ..... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 38 | 62 | 51 | 96 | 49 | 78 | 82 | 91 | 65 |

Determine the median number of people with diabetes.

**(2)**

median = .....

**(Total for Question 7 = 10 marks)**

## 7. Question 7 · part b · 1 mark · Working and answer

Working and answer

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# 8. Question 3 · part c.i · 1 mark

June 2023 Paper 2

Further practice · Attempt part c.i only.

**3** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

**(2)**

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(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

**(3)**

number of people = ..... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

**(2)**

number of people = ..... million

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P 7 2 5 8 7 A 0 6 2 4

## 8. Question 3 · part c.i · 1 mark

June 2023 Paper 2

Further practice · Attempt part c.i only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

**(1)**

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

**(2)**

(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

**(3)**

**(Total for Question 3 = 13 marks)**

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## 8. Question 3 · part c.i · 1 mark · Working and answer

Working and answer

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## 9. Question 4 · part a.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.ii only.

**4** Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

**(2)**

(ii) Name the hormone that stimulates the liver to produce glycogen.

**(1)**

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

**(1)**

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

| BMI | Category |
|---|---|
| less than 18.5 | underweight |
| 18.5 to 24.9 | healthy weight |
| 25 to 29.9 | overweight |
| 30 to 34.9 | obese |
| 35 to 39.9 | severely obese |
| more than 39.9 | morbidly obese |

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## 9. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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## 10. Question 4 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

**4** Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

**(2)**

(ii) Name the hormone that stimulates the liver to produce glycogen.

**(1)**

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

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## 10. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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# 11. Question 3 · part c.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part c.ii only.

**3** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

**(2)**

.....

.....

.....

.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

**(3)**

number of people = ..... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

**(2)**

number of people = ..... million

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## 11. Question 3 · part c.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part c.ii only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

**(1)**

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

**(2)**

(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

**(3)**

**(Total for Question 3 = 13 marks)**

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## 11. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

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